PSS ACCEPTANCE 360

F02 | Technical, Integration, Cybersecurity & Certification Feasibility Assessment

Architecture evidence, proof-of-capability methodology, certification scope, security and deployment control gates

Prepared for prospective Research, Product & Business Development programme | 19 October 2026 | Version 1.0

DECISION STATUS - FEASIBILITY NOT DEMONSTRATED

A read-only, bank-scoped PSS Operate + Connect service is technically plausible as a DESIGN HYPOTHESIS. No internal PSS infrastructure audit, bank interface demonstration, penetration test, certification verification, technical costing or independent CISO sign-off has occurred. This document supports a bounded evidence-gathering mandate only.

Document control and interpretation

Item	Controlled position
Reference	PSS-A360-F02-001 / Volume III; companion evidence-and-test workbook F02-R01.
Purpose	Assess technical plausibility, enumerate disqualifying gaps, specify verifiable acceptance tests and estimate research prerequisites.
Author / authority	Independent pre-appointment working draft; not authorised by or issued on behalf of PSS.
Evidence baseline	PSS project drafts S01-S04; V02-V05; P01-P04 in the private research package; publicly available official standards/law checked 19 Oct 2026.
Not examined	Actual network diagrams, configurations, logs, source code, device apps, merchant/settlement data, vendor contracts, PCI attestations or bank authorisations.
Approvals pending	PSS CTO, Head of Operations, CISO/Risk, Legal/Compliance, Finance, participating bank and applicable regulator/scheme/supplier.
Document status	DRAFT - desk-level conditional plausibility, technical validation pending.

Executive feasibility finding

The first potential product is a bank-authorised observational merchant operations layer. Under P01 and P02, PSS would ingest permitted evidence from authoritative sources, maintain versioned source-to-event mappings, distinguish uncertain payment outcomes, reconcile reports and support exception cases. The architecture deliberately avoids the initiation of financial transfers, settlement instructions, bank ledger posting and unilateral merchant refunds. P03 proposes bank/PSS operating duties; P04 retains competitive build-buy-partner options. None is an authorised production operating model.

Dimension	Desk judgement	Critical unknown
Architecture modularity	Conditionally feasible in principle	Can existing PSS systems expose stable permissioned event and settlement feeds?
Payment semantics	Technically specifiable	Are source status codes complete, authoritative and independently reconcilable?
Bank tenancy / access	High-risk but testable	Can per-bank access be enforced end-to-end including exports, caches and backups?
Cybersecurity / resilience	Not assessed in PSS environment	Scope, controls, security testing, SOC, RTO/RPO and incident capability.
Certifications and permissions	Not verified	Active PCI DSS AoC scope, any relevant MPoC listing, scheme and national-rail terms.
Hosting / lawful data use	Undetermined	Locations, vendor constraints, personal data processing roles and authorised sharing.
Cost / capacity	Unquantified	Infrastructure headroom, internal staffing, support, cyber controls

Dimension	Desk judgement	Critical unknown
		and supplier fees.

MANAGEMENT RECOMMENDATION

Proceed only to authorised S03/V05 verification, architecture discovery and synthetic lab demonstrations. Do not approve a live bank data feed, merchant production launch, supplier award or certification claim on the basis of this F02 draft.

1. Scope and exclusions

In scope for candidate release R0: bank-scoped ingestion from explicitly authorised feeds; immutable raw-source evidence pointers and integrity checks; idempotent normalisation; four independent status dimensions; bank-specific reconciliation; exception queues; bank operations workspace; audit, monitoring, security and reporting. Conditional channels (SoftPOS, POS, gateway, EthQR, wallet) are considered only as potential event sources under their own agreements.

Out of scope: a new payment switch or QR scheme, payment initiation, direct access to bank funds, payment authorisation decisions, KYC approval, final settlement decisions, ledger writes, autonomous refund or credit decisions, consumer scoring, a shared-agent rollout and production deployment. PSS Secure / 3-D Secure has a separate card-not-present authentication purpose.

2. Source documents and design inheritance

Reference	Inherited design requirement	F02 verification concern
S03 / V05	Reported PSS switching, POS, PSS Pay, gateway and security assets are NOT internally verified.	Document/interface inventory; runtime logs and current certifications.
P01	40 proposed functional and 14 quality requirements; first target is Operate + Connect.	System-specific testability, interface fidelity and service costs.
P02	12 candidate components/interfaces; canonical evidence envelope; independent payment/reversal/recon/settlement states.	Source accuracy, replay, partial failures and tenant boundaries.
P03	Bank-sponsored operational model; proposed controls, RACI and cases.	Service owner, handoff proof, bank sign-off and financial authority.
P04	Reuse-first modular hybrid is a sourcing hypothesis, not a winner.	Demonstrate build/buy viability and validated supplier security posture.
V03/V04	Bank demand and merchant benefit still unproven.	Do not overengineer before credible sponsor/pricing evidence.

3. Reference logical architecture

Proposed flow: (1) authorised bank/PSS source event or settlement report; (2) bank-scoped contract adapter and intake quarantine; (3) durable source record with timestamp and provenance; (4) schema/status mapping with reject-or-unknown default; (5) reconciliation and exception engine; (6) role-gated bank operational portal/export; (7) monitoring, immutable audit and evidence retention controls. Any action affecting financial rights is routed to the authorised bank under P03, not executed by the observational service.

Layer	Design test	Decision dependency
Ingress and identity	Source authentication, purpose/tenant scoping, message signature or transport integrity, input validation.	Real interface permission and specs.
Evidence journal	Append-only semantic history; raw-status preservation; checksum/provenance; clock skew handling.	Retention and tamper evidence; lawful personal data minimisation.
Transformation	Versioned mappings; unknown as default; compatibility;	Authoritative source dictionary and test

Layer	Design test	Decision dependency
	duplicate and conflict detection.	vectors.
Matching and cases	Independent bank statements; explainable matching; unmatched queues; no silent adjustments.	Bank finance reconciliation rulebook.
Presentation and export	Least privilege; masked data; scoped search; signed audit trail; no unsourced success display.	IAM integration and privacy approvals.
Security / availability	Network zones, secrets, encryption, telemetry, patching, backup/restore, SOC response.	Validated PSS CISO architecture and hosting choice.

4. Four independent transaction-status dimensions

Dimension	Allowed evidence	Forbidden shortcut
Payment outcome	Competent processing party confirms/declines/unknown; source status/time retained.	Assume success from API acknowledgement, timeout or prior authorisation.
Reversal / refund	Original reference plus separate source-authorised reversal/remedy confirmation.	Rewrite original success event or create unilateral refund.
Reconciliation	Bank/scheme/PSS records matched under an approved rule and cutoff.	Equate a processing event with ledger reconciliation.
Settlement proof	Qualified bank/settlement-party statement, amount, date and reference.	Show paid/settled because a transaction was accepted or matched.

The implementation must support pending, indeterminate, conflicting and late-arriving events without moving an outcome to “successful” by inference. All changes require versioned mapping logic and recorded authorship. Bank-finance evidence retains precedence for posting/settlement confirmation.

5. Integration feasibility and boundary map

Candidate dependency	Required proof	Current status
Bank merchant / account registry	Schema, bank mandate, authoritative identifiers, role of KYC states.	Not inspected
Bank ledger / settlement records	Signed reference, cutoff, settlement vs processing semantics and consent.	Not inspected
PSS switch / POS event feed	Coverage, latency, reliability, reversals and ownership of event data.	Not inspected
PSS Pay / SoftPOS	Supported OS/device, acquirer, exact PCI/scheme scope, permitted data output.	Not inspected
PSS payment gateway	Actual production capabilities, event status, contract and PCI scope.	Not inspected
EthSwitch / EthQR	Authorised participant route and scheme contract; event compatibility.	Not inspected
Wallet routes	Provider/partner consent, statement reconciliation and pricing.	Not inspected
Bank/PSS IAM, service desk	Federation, separation of duties, case handoff and customer communications.	Not inspected

6. Cybersecurity feasibility framework

Use NIST CSF 2.0 to organise governance, identifying exposures, protection, detection, incident response and recovery [EX-01]. Apply NIST SP 800-218 v1.1 (final) as an engineering and third-party software evidence vocabulary [EX-02]. Revision 1 / v1.2 is a public draft, not assumed final [EX-03]. Apply OWASP API Security Top 10 (2023) to bank adapter interfaces and portal APIs [EX-04]. These are reference practices, not proof of compliance in PSS systems.

Control family	Expected evidence and required adversarial test
Identity / privilege	Service-to-service authentication and scoped roles, MFA for privileged staff, joiner/mover/leaver

Control family	Expected evidence and required adversarial test
	records; denied cross-bank object and property reads.
Tenant segregation	Adversarial bank A/B tests across APIs, case queues, search, reports, analytics, caches, exports, backups and support access.
Integrity / replay	Event digests, source-message identity, duplicate and conflicting updates, poisoned events, out-of-order delivery, tamper alerts.
Confidentiality / privacy	Field inventory; minimised/tokenised identifiers; redaction; lawful purpose, retention, location and access; no raw PAN/PIN unless separately approved scope.
Secure software / vendor chain	Repository permissions, threat model, code review, dependency and supply-chain controls, SBOM where relevant, vulnerability SLA, third-party attestations.
Monitoring and incident	Centralised audit/metrics, event source lag, SIEM/SOC workflow, escalation authority and breach/incident notifications.
Availability / restoration	Fail-safe queueing, backups, measured recovery time/data loss, restored audit provenance and idempotent replay.
Operational governance	Four-eye changes for status mappings and financial-case overrides; trusted-source contract changes and controlled releases.

7. Certification, regulatory and scheme scope

Instrument / authority	What is known publicly	F02 condition
PCI SSC MPoC	PCI SSC defines MPoC for eligible contactless/PIN acceptance on COTS devices [EX-05].	Only applicable if PSS Accept/SoftPOS is in scope: verify exact listed solution/software/service, OS, acquirer and scheme requirements; no inference from general PCI DSS.
PCI DSS	PCI security controls depend on actual cardholder data environment and service scope.	Request current signed AoC/ROC scope and service-provider responsibility matrix; no claim of coverage for new design.
EthQR / national rails	Interoperable rails exist within designated participants and rules.	Confirm contractual scheme membership, participant roles, technical permissions and message duties.
NBE policy	NBE web page still labels NDPS 2026-30 as draft [EX-06].	Do not use draft action milestones as operative regulatory permission.
Ethiopia PDPA	Ministry of Justice lists Proclamation 1321/2024 in force [EX-07].	Legal to determine controller/processor and lawful basis; sharing, retention, storage geography and cross-border requirements.
Actual PSS licences	Earlier S03 notes company-reported POS/gateway permissions but no original licence files.	Legal must inspect grant instruments, limitations, notification/outsourcing restrictions and operator accountabilities.

Other frameworks (e.g. ISO 20022) may help data dictionary compatibility where required by a bank or scheme, but no specific ISO 20022 message implementation is assumed or demanded without source-interface evidence. Scheme certification, PCI DSS and MPoC listing are distinct forms of evidence.

8. Risk and threat hypotheses - priority blockers

ID	Failure hypothesis	Mitigation and test / owner
TH-01	One bank can retrieve another bank's transactions.	Tenant-escape penetration test on API, query, cache, batch and report paths; CISO.
TH-02	Unknown/timeout payment is represented as paid.	Indeterminate default; signed-source tests; bank Ops.
TH-03	Replay leads to duplicated cases or a false financial instruction.	Idempotency-key collision/conflict and recovery tests; Technology.
TH-04	Reconciliation mismatches disappear in aggregation.	Full exception export and cut-off parity against authorised bank records; Bank Finance.
TH-05	Compromised adapter supplies plausible forged source statuses.	mTLS/signature verification, source catalogue, quarantine, abuse tests; CISO.
TH-06	PAN/PII traverses a system outside allowed scope.	Data lineage, sample payload inspection, deletion, PCI/legal analysis;

ID	Failure hypothesis	Mitigation and test / owner
		Legal/CISO.
TH-07	Vendor or platform cannot recover timely after outage.	Restore and site-failure rehearsals with measured RTO/RPO; PSS Ops.
TH-08	SoftPOS certification/OS mismatch invalidates acceptance channel.	Solution listing and contract/scheme validation; acquirer/PCI specialist.
TH-09	Privileged user falsifies the status map or closes an adverse case.	Dual-control adversarial test and immutable change trail; Risk.
TH-10	Contract excludes needed data rights or ownership.	Legal/Procurement licence review and exit-data export test.

9. Evaluation protocol and scoring rules

F02 feasibility is not based on a single arbitrary numeric score. Apply a four-part decision procedure: (a) disqualifying legal/security gates; (b) requirement-based demonstrated capability; (c) measured reliability/recovery/performance against a justified workload; and (d) documented remediation cost/dependencies. An unexecuted test is OPEN, not a pass. “Not applicable” must carry reviewer justification.

Test state	Meaning	Allowed decision use
OPEN	No authorised execution or admissible evidence.	Counts as evidence gap, never success.
IN REVIEW	Evidence submitted but unchecked by appropriate reviewer.	No gate release.
PASS	Reproducible artefact and acceptance criteria met with signed reviewer.	Supports specified requirement only.
FAIL	Observed deviation or unmet criterion.	Remediate/retest or stop.
BLOCKED	Test cannot proceed due to permission, environment, vendor or missing source.	Escalate; no positive inference.
N/A	Reviewer documents why item is outside the approved release scope.	Excluded with traceability, not suppressed.

Proposed indicative measures to CALIBRATE, not approve as SLAs: source-to-ingestion lag percentile; reconciliation coverage by qualified source and cutoff; unresolved-unknown ageing; duplicate-event detection; missed exceptions; tenant-escape critical defect count; batch/file completeness; restore time and data loss; case backlog; authorised mapping change auditability. Baselines and thresholds shall be set after representative source workload analysis and bank-specific terms.

10. Technical demonstration test plan

Phase	Test package	Minimum observable output
A / desk - now	Contract templates, schemas, threat analysis, synthetic event corpus, certification request inventory.	Reusable test protocol; ZERO claims of live validation.
B / system discovery - authorised	PSS architecture walkthrough; extract controlled configuration; sample data dictionary; licence/AoC scope.	Signed asset inventory, source permissions and initial fit-gap.
C / isolated PoC - authorised	Synthetic two-bank integration, duplicate and unknown events, late reversals, matching, IAM and fault tests.	Timestamped reproducible logs, test results, vulnerability/defect list.
D / controlled bank sandbox	Partner-approved status/settlement evidence feeds, service desks, realistic cutoffs and volume.	Independent reconciliation sample; bank finance, Ops and CISO sign-off.
E / investment gate	Consolidate findings, remediation costs, residual threats, partner constraints, sourcing comparison.	Go / revise / defer / stop; no automatic live-pilot approval.

11. Hard technical and institutional gates

The following must be PASS with accountable evidence, or formally excluded where truly not in scope, before a live pilot recommendation: verified PSS licence boundaries; approved bank agreements and data-sharing purpose;

current scoped PCI/scheme evidence for every activated channel; zero critical tenant-boundary failures; zero optimistic payment/settlement status defects; independently reproducible reconciliation; tested replay and recovery; vulnerability management and penetration-test resolution; bank/PSS roles and incident routing; privacy sign-off; costed hosting/support model; named accountable sponsor; approved pilot environment and change controls. The workbook provides fourteen individually owned hard gates with fields for evidence and sign-off.

12. Cost and deployment feasibility input requirements

Cost element	Evidence to request	Model linkage
Integration	PSS/bank adapter build, test, certification and change cost by bank.	F05 CAPEX / unit onboarding cost.
Runtime	Compute, storage, networking, security controls, logs, backups, DR region/location.	F05 monthly fixed and variable OPEX.
Data / retained evidence	Event volume, retention, masking, lawful storage location and rehydration.	Cost per event and per active merchant.
People / support	24x7 vs business-hour operation, L1-L3 staffing, bank support desk and SOC.	Headcount, operations and incident response.
Supplier / licences	Third-party contracts, per-instance/transaction fees, FX exposure, exit rights.	P04 TCO comparison and F05 sensitivities.
Certification	PCI/scheme vendor and independent assessment cost for conditional ACCEPT.	Separate gated investment item.

No CAPEX, OPEX, workload capacity, annual availability percentage, disaster recovery RTO/RPO or technical ROI is verified for PSS by this report. No invented score or cost is included.

13. Accountabilities and independent challenge

Function	Required F02 contribution / sign-off
Research / Product	Maintain requirement-to-test matrix, use cases, exclusions and business relevance.
PSS Technology / Architecture	Asset/interface proof, technical compatibility, capacity and implementation estimate.
PSS Operations	Workflow controls, monitoring, incident and recovery drill results.
PSS CISO / Risk	Independent threat model, vulnerability and tenant-isolation test acceptance, residual risk.
PSS Legal / Compliance	Legal basis, licence/outsourcing and certification/scheme responsibility opinion.
PSS Finance / CFO	Corroborate run costs, staffing, supplier quotes and financial sensitivity inputs.
Sponsoring bank	Formal API rights, KYC/settlement boundary, bank-source samples and finance reconciliation sign-off.
Executive Committee	Separate authorisation for feasibility, pilot readiness and later scale; budget ceilings.

14. Pre-appointment notebook and post-joining evidence requests

Prepare now from public/owned material	Execute only after joining with authority
Finish synthetic bank A/B event fixture library.	Obtain actual PSS network/data-flow and product architecture diagrams.
Build normalisation and duplicate test dictionaries.	Request PSS Pay / POS / gateway event dictionaries and operational logs.
Maintain PCI, NIST, OWASP, PDPA standard crosswalk.	Review current AoC, scheme approvals, SoftPOS listings and licence terms.
Draft permissioned bank data-sharing and API questionnaire.	Authorise lead-bank source sample and reconciliation control test.
Model open/blocked/failed technical gates.	Conduct authorised lab, security and recovery tests and sign results.
Prepare options for reuse-first and no-build baseline.	Obtain supplier quotes, hosting constraints and CFO cost confirmation.

15. Formal conditional technical verdict

F02 DISPOSITION: CONDITIONAL - EVIDENCE NOT YET AVAILABLE

Technical concept feasibility: PLAUSIBLE IN PRINCIPLE. PSS-specific implementability: UNKNOWN. Security posture: NOT TESTED. Certifications, participant permissions, data rights: UNVERIFIED. Capacity and lifecycle cost: UNQUANTIFIED. Recommendation: advance controlled discovery and synthetic proof-of-concept once authority is obtained; do not approve procurement, live pilot or public readiness claims.

Reversal rule: abandon Operate + Connect first if permitted transaction/settlement data cannot be obtained on lawful, reliable and affordable terms; if a regulator or scheme boundary prevents the intended service; if segregation or reconciliation cannot pass independent tests; or if the cost and support burden erases the bank value case. Retain the PSS Pay-led, partner-only and no-build comparators from S02.

Annex A. Official external references and their limits

ID	Official source	Relevance / status
EX-01	https://www.nist.gov/news-events/news/2024/02/nist-releases-version-20-landmark-cybersecurity-framework	NIST CSF 2.0 structure; voluntary security framework, not local licence.
EX-02	https://csrc.nist.gov/pubs/sp/800/218/final	NIST SSDF v1.1 final; secure development evidence vocabulary.
EX-03	https://csrc.nist.gov/pubs/sp/800/218/r1/ipd	SSDF v1.2 initial public draft only as of evidence cut-off.
EX-04	https://api-security.owasp.org/editions/2023/en/0x00-header/	OWASP API Top 10 (2023) security testing reference.
EX-05	https://www.pcisecuritystandards.org/standards/mobile-payments-on-cots-mpoc/	PCI MPoC programme; do not infer an actual product listing.
EX-05a	https://listings.pcisecuritystandards.org/assessors_and_solutions/mpoc_solutions?agree=true	Authoritative MPoC solution listing to verify exact vendor/configuration.
EX-06	https://nbe.gov.et/ndps/	NBE still labels strategy as draft; not operative authority.
EX-07	https://justice.gov.et/en/law/personal-data-protection-proclamation/	Justice Ministry confirms Proclamation 1321/2024 in force.
EX-08	https://www.nist.gov/cyberframework/faqs	Six-function framework and governance definitions.

Document-source distinction: S01-S04, V02-V05 and P01-P04 are privately prepared PSS Acceptance 360 proposals, not externally verified PSS system evidence. The external sources above establish general standards/law, not PSS implementation, approval or compliance.

Annex B. Sign-off record - intentionally blank

Role	Name / date	Decision / evidence ID
PSS Technology	Not assigned	Pending
PSS Operations	Not assigned	Pending
PSS CISO/Risk	Not assigned	Pending
PSS Legal/Compliance	Not assigned	Pending
PSS CFO	Not assigned	Pending
Lead bank	Not assigned	Pending
Programme sponsor	Not assigned	Pending